Deepak Sathyan

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Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University, College Station, TX

Education

University of Maryland — College Park, MD
Ph.D. in Physics — Aug 2018 – July 2024
Thesis: Unifying Searches for New Physics with Precision Measurements of the W Boson Mass
Advisor: Dr. Kaustubh Agashe

Boston University — Boston, MA
Bachelor of Arts in Physics, Summa Cum Laude with Honors — Aug 2014 – May 2018
GPA: 3.9
Minor: Mathematics
Thesis: Muon (g-2): Searching for the Muon Electric Dipole Moment
Advisor: Dr. Robert Carey

Experience

Postdoctoral Research Associate — College Station, TX
Mitchell Institute for Fundamental Physics and Astronomy — Sept 2024 – Aug 2027
Texas A&M University — 3 years

Publications

Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering — May 2026
B. Dutta, D. Goswami, J. Kumar, M. Rai, D. Sathyan
arxiv.org/abs/2605.08010

A Baryon and Lepton Number Violation Model Testable at the LHC — Aug 2025
A. Bhoonah, F. Burk, D. Liu, T. Ou, D. Sathyan
arxiv.org/abs/2508.21064

New Constraints on Neutrino-Dark Matter Interactions: A Comprehensive Analysis — July 2025
P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang
arxiv.org/abs/2507.01000

New laboratory constraints on neutrinophilic mediators — 2025
P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang
arxiv.org/abs/2407.12738 (Phys. Lett. B)

'Unification' of BSM searches and SM measurements: the case of lepton+MET and m_W — 2025
K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
arxiv.org/abs/2404.17574 (JHEP)

A new purpose for the W-boson mass measurement: Searching for New Physics in lepton+MET — 2024
K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
arxiv.org/abs/2310.13687 (Phys. Lett. B)

Unifying Searches for New Physics with Precision Measurements of the W Boson Mass 2024
D. Sathyan
drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9 (Ph.D. Thesis)

Energy-peak based method to measure top quark mass via B-hadron decay lengths 2023
K. Agashe, S. Airen, R. Franceschini, J. Incandela, D. Kim, D. Sathyan
arxiv.org/abs/2212.03929 (JHEP)

TF07 Snowmass Report: Theory of Collider Phenomena Oct 2022
Collaboration
arxiv.org/abs/2210.02591

Report of the Topical Group on Physics Beyond the Standard Model at Energy Frontier for Snowmass 2021 Sept 2022
Collaboration
arxiv.org/abs/2209.13128

Report of the Topical Group on Top quark physics and heavy flavor production for Snowmass 2021 Sept 2022
Collaboration
arxiv.org/abs/2209.11267

Snowmass2021 - White Paper, Implications of Energy Peak for Collider Phenomenology: Top Quark Mass Determination and Beyond Apr 2022
K. Agashe, S. Airen, R. Franceschini, D. Kim, D. Sathyan
arxiv.org/abs/2204.02928

Snowmass2021 White Paper: Collider Physics Opportunities of Extended Warped Extra-Dimensional Models Mar 2022
K. Agashe, J.H. Collins, P. Du, M. Ekhterachian, S. Hong, D. Kim, R.K. Mishra, D. Sathyan
arxiv.org/abs/2203.13305

The straw tracking detector for the Fermilab Muon g-2 Experiment 2022
Collaboration
arxiv.org/abs/2111.02076 (JINST)

Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm 2021
Muon g-2
arxiv.org/abs/2104.03281 (Phys. Rev. Lett.)

Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab 2021
Muon g-2
arxiv.org/abs/2104.03240 (Phys. Rev. Accel. Beams)

LHC Signals for KK Graviton from an Extended Warped Extra Dimension 2020
K. Agashe, M. Ekhterachian, D. Kim, D. Sathyan
arxiv.org/abs/2008.06480 (JHEP)

Seminars

Unification of Searches and Measurements: Probing BSM with the W boson mass measurement
CMS Exotica general meeting, May 2024

A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
Theory Seminar at Washington University in St. Louis, October 2023
HEP Seminar at Northwestern University, October 2023
HEP Seminar at University of Pittsburgh, November 2023
CMS SUSY general meeting, November 2023
Seminar at UT Austin, December 2023
Seminar at Texas A&M, December 2023

Conference Talks

Can the LHC be sensitive to the Light Dark World?
Light Dark World 2026 at Carleton University, July 2026

Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering
CETUP* Workshop, June 2026

A Baryon and Lepton Number Violaton Model Testable at the LHC
Particle Physics on the Plains at the University of Kansas, November 2025

Can the LHC be sensitive to light dark mediators?
Phenomenology Conference at the University of Pittsburgh, May 2025
CETUP* Workshop, June 2025

A comprehensive analysis of supernova neutrino-dark matter interactions
Mitchell Conference at Texas A&M University, May 2024
NuFact Conference, September 2024
Texas TACOS @ UT Austin, October 2024
Particle Physics on the Plains at the University of Kansas, November 2024

A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET
Particle Physics on the Plains at the University of Kansas, October 2023

Probing Dark Matter-Neutrino Interactions via Supernova Neutrinos
Phenomenology Conference at the University of Pittsburgh, May 2023

Model-Independent Measurement of Top Quark Mass Using B-Hadron Decay Lengths (Part I)
Phenomenology Conference at the University of Pittsburgh, May 2022

Signals of KK graviton from extended warped extra dimensions at the LHC (II)
Phenomenology Conference at the University of Pittsburgh, May 2020
April APS Meeting, April 2021

Projects

BSM Searches via W Boson Mass Jan 2020 – July 2024
Developed a framework unifying beyond-the-Standard-Model searches with precision measurements of the W boson mass at the LHC.

Muon g-2 Experiment Aug 2018 – Jan 2022
Contributed to the straw tracking detector and beam dynamics corrections for the Fermilab Muon g-2 experiment, which measured the muon anomalous magnetic moment to 0.46 ppm.

Conferences Organized

The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2026 May 2026 – May 2026

The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2025 May 2025 – May 2025

Awards

Breakthrough Prize in Fundamental Physics Apr 2026
Muon g-2 Collaboration

Skills

High Energy Physics

Computing